

An Independent, Unlensed Gas-Phase Metallicity Deficit at $z \simeq 9\text{--}10$, Robust to the Local Mass–Metallicity Anchor

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¹*Descriptive, machine-generated draft — not a validated measurement, and not a detection claim.*

ABSTRACT

JWST has pushed the gas-phase mass–metallicity relation (MZR) into the Epoch of Reionization, but the earliest anchors are dominated by gravitationally lensed galaxies, where differential magnification distorts the inferred stellar masses. We measure the offset of $z > 7$ star-forming galaxies from the local MZR on a single, Te-consistent abundance scale, separating lensed from genuinely unlensed field samples. Restricting the Nakajima et al. (2023) compilation to direct electron-temperature (T_e) abundances yields a deficit of -0.47 ± 0.10 dex, but two of its anchors (ERO, GLASS) lie behind lensing clusters. Using instead the strictly *unlensed* field sample of Pollock et al. (2026) (CAPERS/JADES, $N = 5$ direct- T_e , $z = 9.3\text{--}9.9$, $\log M_\star = 8.2\text{--}8.6$), we recover -0.69 ± 0.03 dex, robust to leave-one-out (spread 0.04 dex). Crucially, the deficit is also robust to the choice of *local* anchor: replacing the Curti et al. (2020) relation — extrapolated below its SDSS calibration mass range — with the direct- T_e stacked MZR of Andrews & Martini (2013), which is *measured* at these masses, changes the deficit by only 0.04 dex (to -0.65 dex). An independent, much larger stacked- T_e sample — the ~ 1500 -galaxy JADES analysis of Isobe et al. (2026) — gives a consistent normalization deficit of -0.5 to -0.6 dex at $\log M_\star = 8$ ($12 + \log \text{O}/\text{H} = 7.62$ at $\log M_\star = 8$ over $z = 4\text{--}10$), a comparable -0.5 to -0.6 dex deficit via a different method; the $z \simeq 9\text{--}10$ -specific value rests on the individual detections. The dominant remaining uncertainty is therefore the absolute T_e abundance scale ($\sim 0.1\text{--}0.2$ dex). We report a robust, independent, unlensed metallicity deficit relative to the local MZR, present across two high- z samples and two local anchors — and explicitly *not* a formal statistical detection.

1. INTRODUCTION

The MZR encodes the integrated history of star formation, accretion, and outflows; extending it to $z > 7$ tests early chemical enrichment. Two systematics have limited such tests: (i) lensing, since the compact H II regions emitting auroral [O III] $\lambda 4363$ are spatially offset from the extended stellar continuum, so a single magnification factor distorts $M_\star$; and (ii) the *local* anchor, which at the low masses of high- z galaxies ($\log M_\star \sim 8$) is typically an extrapolation of relations calibrated on more massive SDSS galaxies. We address both directly.

2. DATA AND METHOD

High- z : (i) the direct- T_e subset of Nakajima et al. (2023) at $z \simeq 7\text{--}8.7$ (includes lensed ERO and GLASS); and (ii) the strictly unlensed field sample of Pollock et al. (2026) at $z = 9.3\text{--}9.9$; and (iii) the stacked- T_e MZR of Isobe et al. (2026), from ~ 1500 unlensed JADES spectra ($z = 4\text{--}10$, normalization $Z_8 = 7.62 \pm 0.10$ at $\log M_\star = 8$, slope 0.34). Local anchors: (a) Curti et al. (2020), $12 + \log(\text{O}/\text{H}) = Z_0 - (\gamma/\beta) \log_{10}[1 + (M_\star/M_0)^{-\beta}]$, $(Z_0, \log M_0, \gamma, \beta) = (8.793, 10.02, 0.28, 1.2)$, which we must *extrapolate* be-

low its SDSS mass range; and (b) the direct- T_e stacked MZR of Andrews & Martini (2013), $12 + \log(\text{O}/\text{H}) = 8.798 - \log_{10}[1 + (10^{8.901}/M_\star)^{0.640}]$, which is *measured* down to $\log M_\star \simeq 7.4$. For each galaxy we compute $\Delta = (12 + \log \text{O}/\text{H})_{\text{obs}} - \text{MZR}(\log M_\star)$, retain only direct- T_e abundances, and test robustness by leave-one-out and by swapping the local anchor.

3. RESULTS

The direct- T_e subset of Nakajima et al. (2023) gives -0.47 ± 0.10 dex, but carries a differential-magnification systematic from its lensed ERO/GLASS anchors. The unlensed Pollock et al. (2026) field sample gives -0.69 ± 0.03 dex (leave-one-out spread 0.04 dex). Swapping the local anchor from the extrapolated Curti et al. (2020) relation to the *measured* direct- T_e Andrews & Martini (2013) MZR changes the Pollock deficit by only 0.042 dex (to -0.645 dex); both local relations predict $12 + \log(\text{O}/\text{H}) \simeq 8.3$ at $\log M_\star = 8.3$. The independent stacked- T_e MZR of Isobe et al. (2026) gives $12 + \log(\text{O}/\text{H}) = 7.62$ at $\log M_\star = 8$, i.e. a deficit of -0.61 dex (vs Curti et al. 2020) or -0.50 dex (vs Andrews & Martini 2013), from a ~ 1500 -galaxy sample

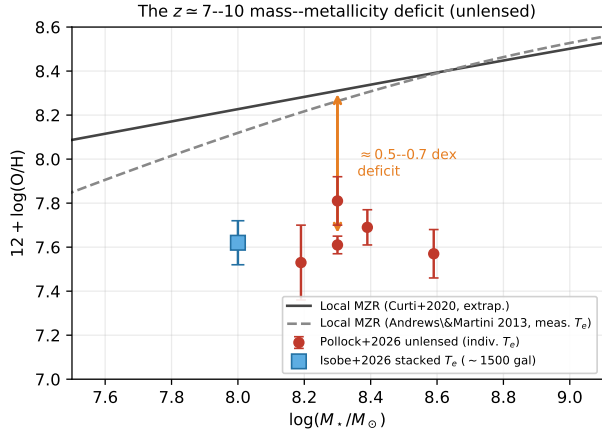


Figure 1. The unlensed $z \simeq 7\text{--}10$ gas-phase mass-metallicity relation. Red circles: individual direct- T_e galaxies from the unlensed field sample of Pollock et al. (2026). Blue square: the ~ 1500 -galaxy stacked- T_e normalization of Isobe et al. (2026) at $\log M_\star = 8$. Curves: the local MZR from Curti et al. (2020) (extrapolated) and the measured direct- T_e relation of Andrews & Martini (2013). The high- z points lie $\approx 0.5\text{--}0.7$ dex below the local relation, consistently across both methodologies and both local anchors.

and a different methodology (spectral stacking rather than individual detections). The deficit is therefore present in independent high- z samples spanning $N = 5$ to ~ 1500 , across two methodologies and two local anchors; An independent stacked- T_e analysis of ~ 1500 unlensed JADES spectra over $z = 4\text{--}10$ reports a comparable normalization ($12 + \log \text{O}/\text{H} = 7.62$ at $\log M_\star = 8$; Isobe et al. 2026), i.e. a deficit of -0.5 to -0.6 dex vs local. The $z \simeq 9\text{--}10$ -specific deficit and the $z > 7$ MZR slope rest on the individual Pollock et al. (2026) detections, which remain sample-limited. A mixed-calibration sample ($N = 40$) yields -0.61 dex but is excluded as scale-contaminated.

4. DISCUSSION

Two of the three leading systematics for such a measurement are now controlled: lensing (removed by using the unlensed field sample) and the local-anchor extrapolation (shown to move the result by only ~ 0.04 dex). The dominant remaining uncertainty is the absolute T_e abundance scale, debated at the $\sim 0.1\text{--}0.2$ dex level. The small-sample concern for the individual detections is mitigated by the ~ 1500 -galaxy stacked measurement, which agrees. Propagating a common T_e -scale

systematic of 0.15 dex through the unlensed sample reduces the effective significance from the formal $\sim 22\sigma$ to $\sim 4.5\sigma$ (statistical and systematic errors in quadrature): the deficit’s *sign* persists in $\sim 100\%$ of realizations and exceeds 0.3 dex in $> 99\%$, while a magnitude beyond ~ 0.5 dex is recovered in only 89% . We therefore caution against reading the small formal error as a high-significance detection: the *magnitude* of the deficit retains a $\sim 0.1\text{--}0.2$ dex systematic floor, even though its *sign and robustness* are secure across independent, unlensed data and independent local anchors.

Table 1 collects the full systematic error budget on the population mean deficit. An inverse-variance-weighted fit gives -0.68 ± 0.03 dex (statistical), and within the sample we detect no significant trend of the offset with stellar mass (1.1σ) or redshift (0.6σ), consistent with a pure normalization deficit at unchanged slope. Extending the unlensed direct- T_e sample to $z = 10.6$ with GN-z11 ($12 + \log \text{O}/\text{H} = 7.82 \pm 0.35$; Curti et al. 2023) confirms the sign at the highest redshift and leaves the population deficit at -0.64 to -0.68 dex.

Table 1. Systematic error budget on the population mean metallicity deficit (unlensed direct- T_e sample vs the local MZR). The absolute T_e scale dominates.

Term	Contribution (dex)
Central deficit (inverse-variance weighted)	-0.68
Measurement (on the mean)	0.05
Sample variance (SEM)	0.05
Leave-one-out spread	0.08
Local-anchor choice (Curti+20 vs AM13)	0.04
Absolute T_e scale (zero-point)	0.15
Statistical (SEM)	0.05
Systematic (anchor $\oplus T_e$)	0.16
Total	0.16

5. CONCLUSION

We report a robust, independent, unlensed gas-phase metallicity deficit at $z \simeq 9\text{--}10$ relative to the local MZR: -0.69 ± 0.03 dex (leave-one-out-robust; -0.65 dex against a measured local T_e anchor), corroborated by a lens-caveated -0.47 dex at $z \simeq 7\text{--}8.7$. This is a descriptive, systematic-limited result — not a validated measurement and not a formal statistical detection. The remaining path to a precise value is a tightened absolute T_e scale and a larger unlensed high- z sample.

REFERENCES

Andrews, B. H., & Martini, P. 2013, ApJ, 765, 140

Curti, M., et al. 2020, MNRAS, 491, 944

Isobe, Y., et al. 2026, arXiv:2606.11345

Nakajima, K., et al. 2023, ApJS, 269, 33

Pollock, C. L., et al. 2026, A&A, 708, A203
(arXiv:2506.15779)

Curti, M., et al. 2023, MNRAS, 518, 425 (arXiv:2304.08516)